

Professional Certificate Programme in Agentic AI and Applications



Fundamentals of AI & ML



Revisit the Learning So Far

Let's take a quick look at the key ideas and concepts you have explored in your learning journey.

- How Python handles errors and exceptions to ensure stable, predictable programme behaviour
- How standard and third-party Python libraries extend functionality and support real-world applications
- How modules are imported to organise code and enable reuse across different scripts
- The role of Python's standard libraries in simplifying common programming tasks
- How to build a command-line calculator using modular code and Python fundamentals

Revisit the Learning So Far

Let's take a quick look at the key ideas and concepts you have explored in your learning journey.

- How to structure a Python project for clarity, scalability and collaboration
- How to work with NumPy and Pandas to manage, analyse and transform data efficiently
- How Python supports data analysis and scientific computing workflows
- How to create data visualisations using Matplotlib and Seaborn for clear insight communication
- How to master code modularisation to improve organisation, reuse and long-term maintainability

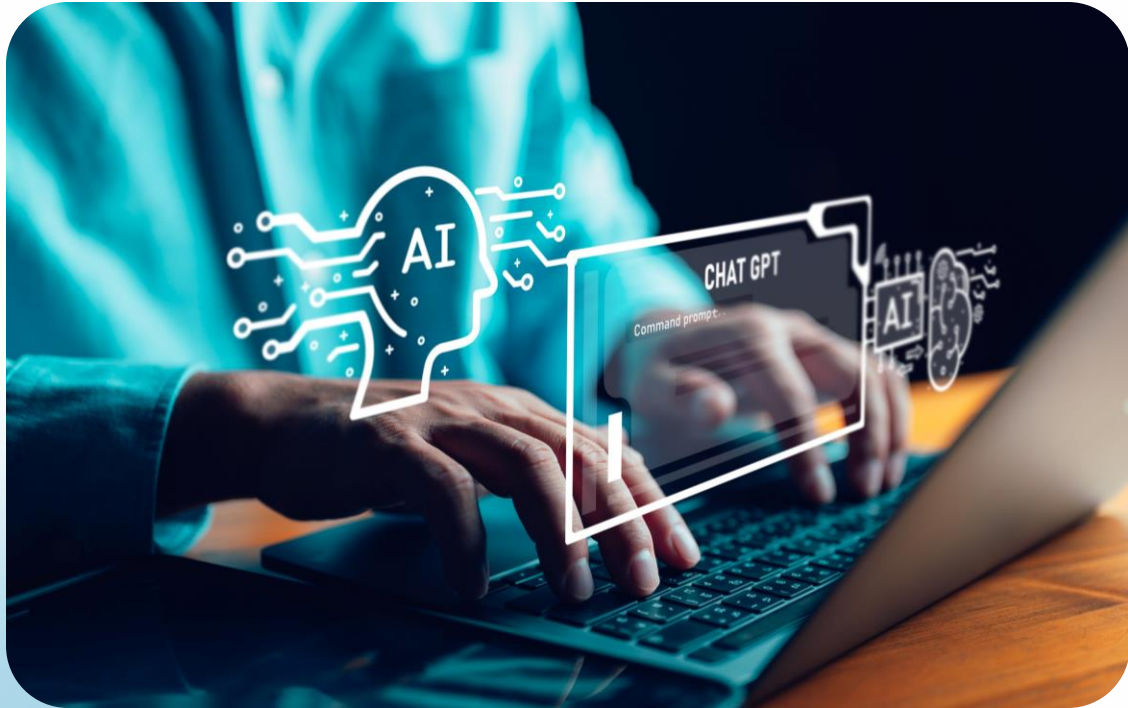
Topics Covered

- AI v. ML v. DL
- Machine Learning
- Types of Machine Learning
- Deep Learning – Neural Networks Introduction
- Reinforcement Learning

AI v. ML v. DL

Artificial Intelligence (AI)

Machines Performing Human-Like Tasks

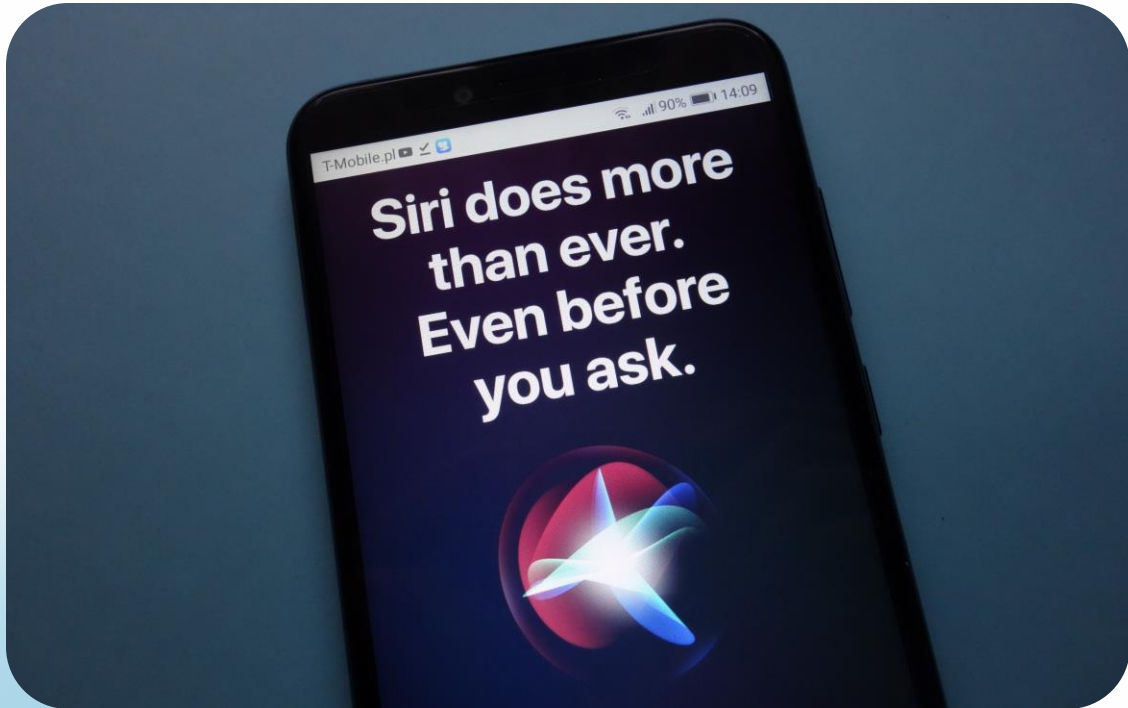


Key capabilities involve:

- Perform reasoning and solve problems
- Perceive and interpret the environment
- Learn from experience and adapt behaviour
- Make decisions under uncertainty

Artificial Intelligence (AI)

Example: Siri



Apple's virtual assistant interprets natural language and performs tasks across apps.

Machine Learning (ML)

Overview



Definition

AI subset where systems learn patterns from data without explicit rules

Key capabilities

Perform predictive modelling, pattern recognition, anomaly detection and classification

Learning process

Improve performance and refine predictions as more data is processed

Machine Learning (ML)

Example: Email Spam Detection



Email systems like Gmail use Machine Learning to automatically identify and filter spam messages based on patterns in message content, sender behavior, and user feedback.

Deep Learning (DL)

Multi-Layer Neural Network Modelling



Key capabilities involve:

- Perform advanced image and facial recognition
- Enable natural language understanding and generation
- Detect complex patterns in unstructured data
- Extract features autonomously without manual engineering

Deep Learning (DL)

Example: ChatGPT

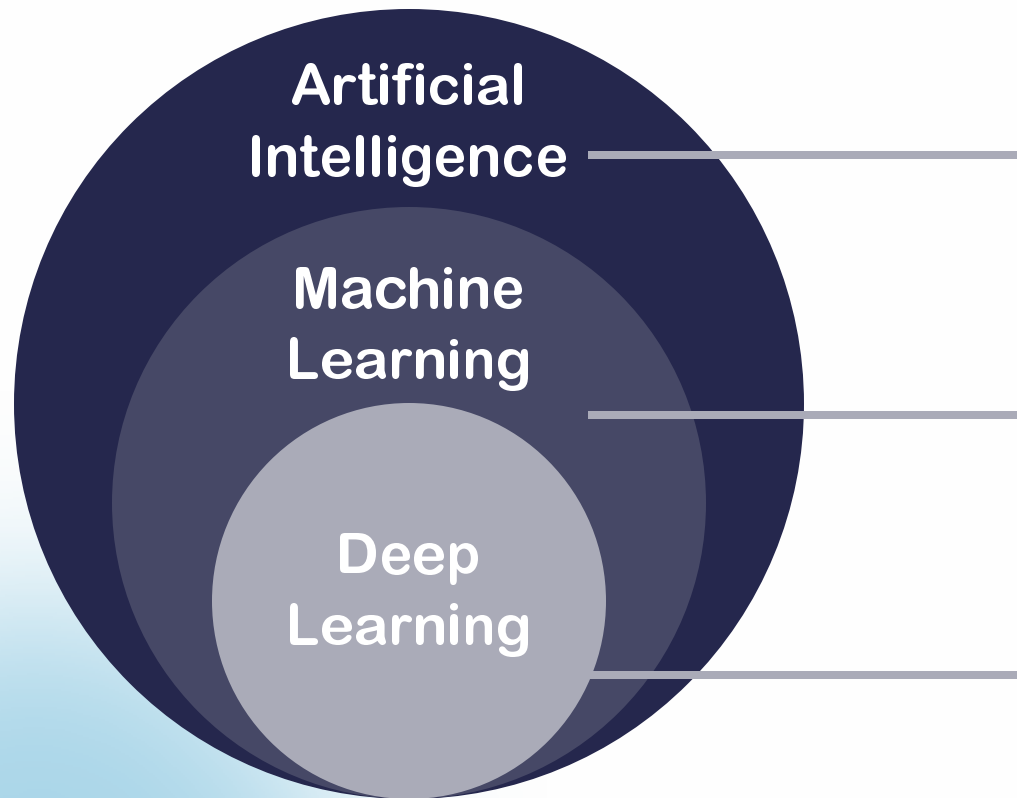


Understand context using
transformer architecture

Generate human-like text and
answer complex questions

Create content across diverse
domains and styles

The AI Family Hierarchy



Artificial Intelligence

Systems that mimic human intelligence, including expert systems and computer vision

Machine Learning

AI subset that learns patterns from data and improves over time

Deep Learning

Multi-layer neural networks modelling complex data for tasks like image recognition and NLP

Comparing AI, ML and DL

Feature	AI	ML	DL
Scope	Broad: Any intelligent system	Focused: Systems that learn	Narrow: Neural-network-based learning
Data needs	Varies	Medium to High	High to Very High (millions of examples)
Processing power	Varies	Moderate to high	Extremely High (often GPU-accelerated)
Human intervention	Varies	Medium to High (feature engineering)	Low to Medium (automatic feature extraction)
Example	Virtual Pas, Autonomous Vehicles, Smart Devices	Recommendation System, Stock Price Prediction	LLM, Language Translation

Modern systems often combine AI, ML and deep learning to optimise performance for complex tasks.

Discussion: AI in Your Daily Life

Where do you encounter AI, ML or DL in your daily life?

Discussion: AI in Your Daily Life

Where Do You Encounter AI, ML or DL in Your Daily Life?

- Netflix recommendations
- Spotify playlists
- Social media feeds

Entertainment



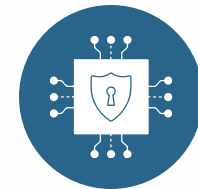
- Email spam filters
- Voice assistants
- Predictive text

Productivity



- Facial recognition
- Fraud detection
- Smart home devices

Security & convenience



Identify which examples use general AI versus specific ML or DL approaches.

Machine Learning

Traditional Programming v. Machine Learning

Traditional programming	Machine learning approach
Provide rules and data	Provide data and outcomes
Follow explicit instructions	Learn patterns automatically
Make logic-based decisions	Adapt to new information
Use rigid rule frameworks	Handle complex relationships

Real-World Applications of Machine Learning



Healthcare

Predict diseases and analyse medical images



Finance

Detect fraud in real-time transactions



Autonomous vehicles

Enhance transportation safety and efficiency



E-commerce

Power recommendation engines driving significant revenue

Types of Machine Learning

Types of Machine Learning



Supervised learning

Train on labeled data to make predictions



Unsupervised learning

Discover hidden patterns in unlabeled data



Semi-supervised learning

Combine labeled and unlabeled data for training



Reinforcement learning

Learn via trial and error with a rewards system

Supervised Learning



Training data

Uses input-output pairs to teach the model

Learning process

Maps inputs to outputs while minimising errors

Prediction

Generates outputs for new data based on learned patterns

Supervised Learning

Examples

Example	Features (Inputs)	Label (Answer)
Email Spam Detection	Email Text: "You won a prize!"	Spam
	Email Text: "Meeting at 10am"	Not Spam
House Price Prediction	Bedrooms: 3, Sq. Ft.: 1500	₹50,00,000
	Bedrooms: 4, Sq. Ft.: 2200	₹75,00,000

Applications of Supervised Learning



Fraud detection (Finance)

Detect fraudulent transactions in real-time



Disease prediction (Healthcare)

Predict diseases from patient history and test results



Customer churn prediction

Identify customers likely to leave a service



Email spam detection

Classify emails using labeled spam and non-spam data

Supervised Learning Algorithms



Classification

- Logistic regression
- Decision trees
- Random forest
- Support vector machines



Regression

- Linear regression
- Polynomial regression
- Gradient boosting
- Neural networks

Unsupervised Learning



Requires no labels

Analyses raw data without predefined categories

Discovers patterns

Identifies natural groupings and relationships

Unsupervised Learning

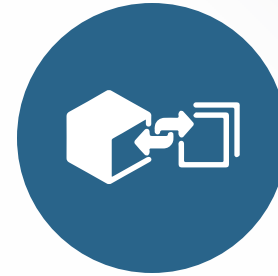
Common Applications



Customer
segmentation



Anomaly
detection



Dimensionality
reduction

Unsupervised Learning

Example

Customer ID	Age	City	Monthly Spending (₹)	Purchase Frequency (per month)
001	27	Hyderabad	8,500	5
002	45	Mumbai	12,000	2
003	33	Bangalore	7,200	3
004	54	Delhi	4,500	1

Applications of Unsupervised Learning

Customer segmentation

Group customers by purchasing behaviour



Anomaly detection

Identify unusual patterns in network traffic



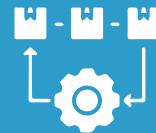
Product recommendation

Suggest products using clustering and association rules



Gene sequence clustering

Group genes by similarity to detect disease-causing mutations



Unsupervised Learning Algorithms

- K-Means
- DBSCAN
- Hierarchical clustering

Clustering



- Principal Component Analysis
- t-SNE
- Autoencoders

Dimensionality reduction



- Apriori Algorithm
- FP-Growth
- Collaborative filtering

Association



Semi-Supervised Learning

Definition



Uses a small set of labelled data plus large unlabeled data to enhance model performance

Semi-Supervised Learning

Key Characteristics



Bridges
supervised and
unsupervised
approaches



Reduces costs
when labeling
data is expensive



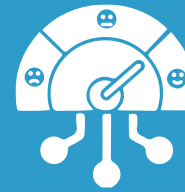
Enables
applications in
NLP, image
recognition and
fraud detection

Applications of Semi-Supervised Learning



Medical image analysis

Detect tumors from MRI/X-ray images using mixed datasets



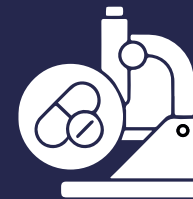
Sentiment analysis

Analyse customer emotions from limited labelled reviews



Fake news detection

Classify unverified articles using verified examples



Drug discovery

Identify new compounds with limited labeled and large unlabelled data

Semi-Supervised Learning Algorithms

BERT & LSTM

Leverage labelled and unlabelled text for language tasks

Self-training

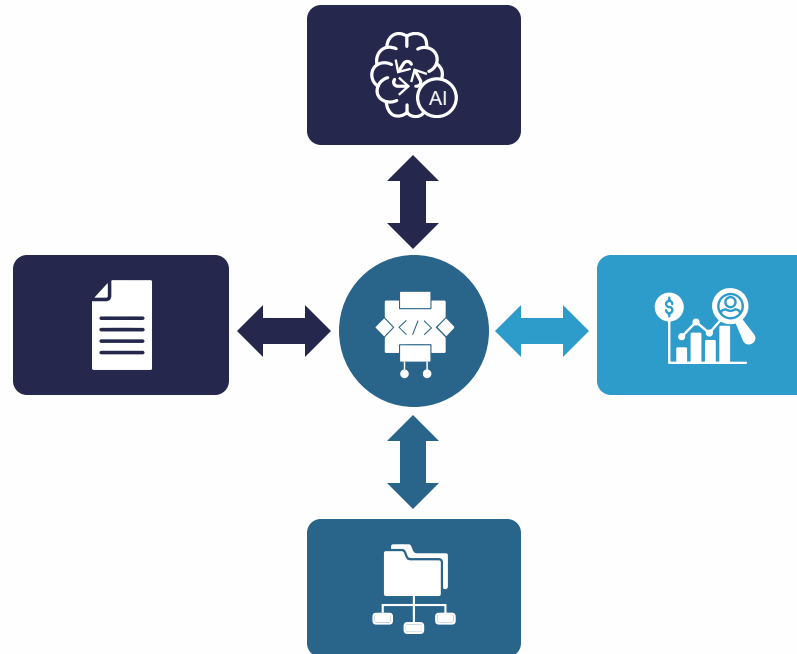
Improve model using its predictions on unlabelled data

Graph-based models

Propagate labels through connected data points

Variational autoencoders

Learn data representations for classification tasks



Reinforcement Learning



Agent

Learner that
takes actions



Action

Choices made
by the agent



Reward

Feedback
guiding the
agent's learning



Environment

World in which
the agent
operates

Notable examples include AlphaGo beating world champions and self-driving car systems.

Applications of Reinforcement Learning



Self-driving cars

Navigate traffic using environmental rewards and penalties



Robotics automation

Optimise industrial robot movements through trial and error



Stock market trading

Learn trading strategies by maximising portfolio returns



Game AI

Master games through millions of simulations

Reinforcement Learning Algorithms



Deep Q-learning

Combine
Q-learning with
neural networks



Policy gradient methods

Optimise policy
performance
directly



Actor-critic models

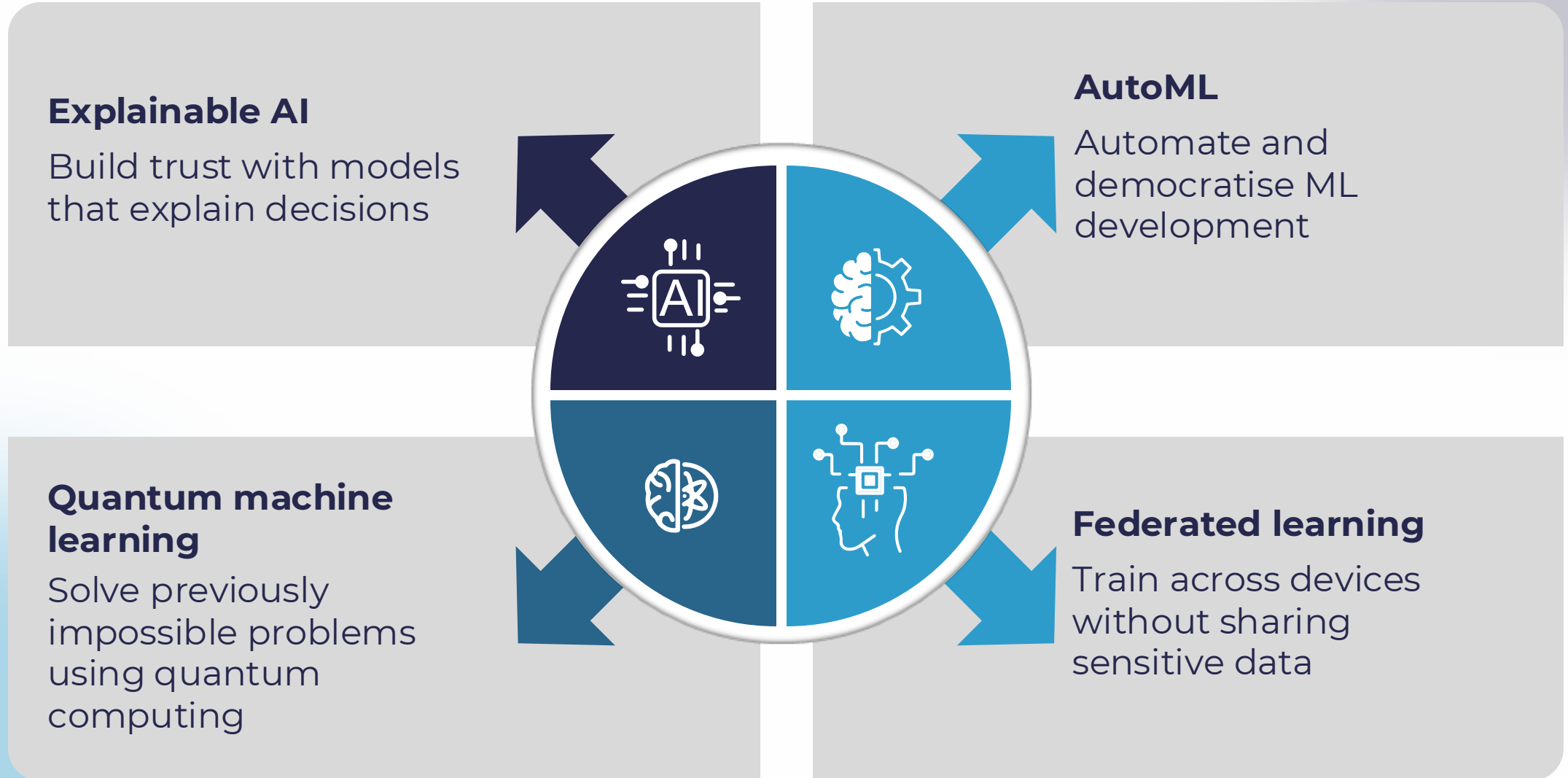
Integrate value-
based and
policy-based
approaches



Proximal policy optimisation

Stable policy
updates for
complex
environments

The Future of Machine Learning



Which ML Type Applies Here?

Case Scenarios



Case 1

A bank wants to predict whether a customer will default on a loan using their income, credit score and spending history.



Case 2

An e-commerce platform wants to segment customers by buying behaviour for personalised marketing.

Which ML Type Applies Here?

Case Scenarios



Case 3

A self-driving car system learns from environmental interactions to optimise driving decisions.



Case 4

A hospital wants to predict patient length of stay using admission data.

Deep Learning – Neural Networks Introduction

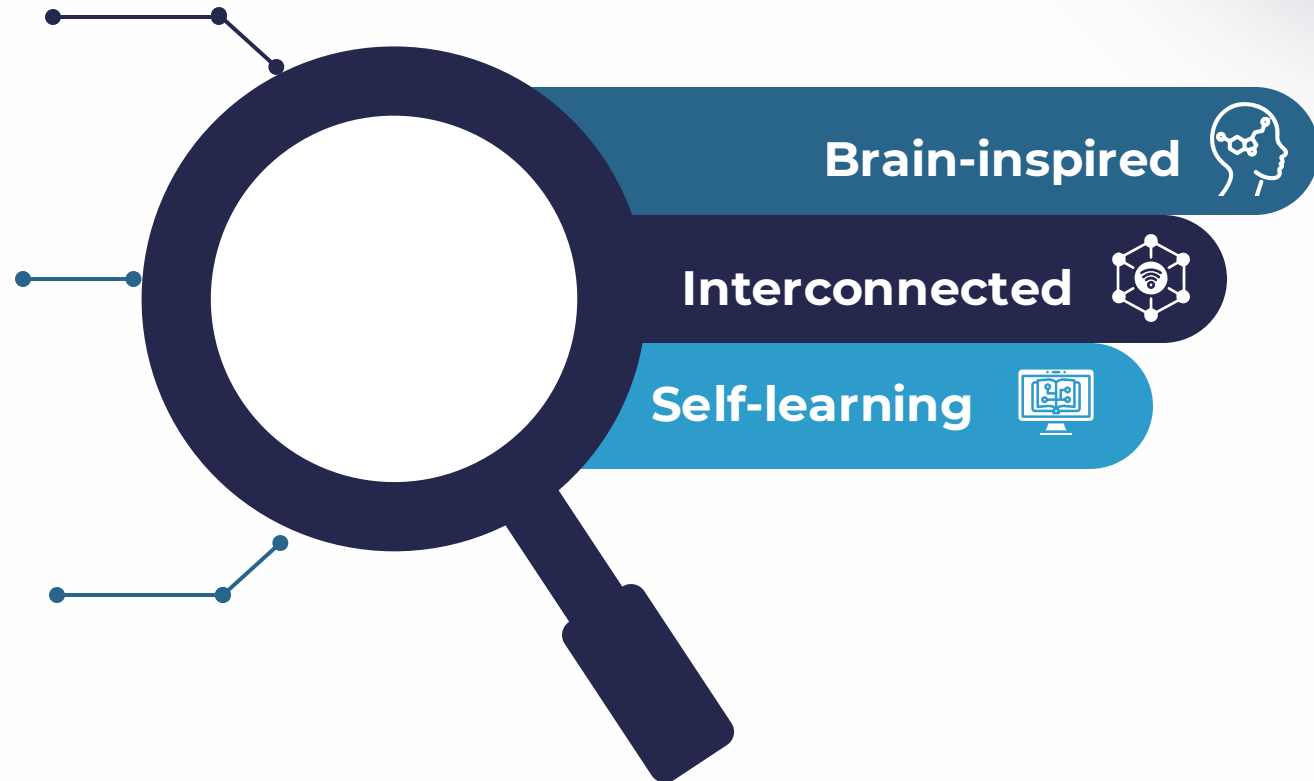
Neural Network

Backbone of Modern Deep Learning

Mimics human brain structure to process information

Connects nodes (neurons) via weighted links

Discovers patterns by adjusting weights and biases



Nodes (Artificial Neurons)

Core Unit of Neural Networks



Input:

Receives features $(X_1, X_2, \dots, X_n)$ from data or previous layer

Weighted sum:

$s = W_1X_1 + W_2X_2 + \dots + b$ Weights (w) determine importance of each input

Activation:

Applies non-linear transformation to the weighted sum

Output:

Passes transformed value to the next layer

Layers in a Neural Network



Input layer

Receives raw features with one node per feature



Hidden layers

Extract complex patterns; deep networks have multiple layers

Deep networks = multiple hidden layers



Output layer

Generates predictions; output shape matches task requirements

Activation Functions

Sigmoid

- Compresses output between 0 and 1
- Used in binary classification output layers

$$f(z) = \frac{1}{1 + e^{-z}}$$

Tanh

- Compresses output between -1 and 1
- Zero-centered, often better for hidden layers

$$f(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

Activation Functions

ReLU

- Returns 0 for negative inputs, input value for positive
- Most popular for hidden layers due to efficiency

$$f(z) = \max(0, z)$$

Softmax

- Converts outputs into probabilities that sum to 1
- Standard for multi-class classification outputs

$$f(Z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Importance of Activations

Prevent layers from collapsing into a single linear transformation

Capture real-world non-linear relationships in data

Allow layers to learn hierarchical, increasingly complex representations

Enabling complexity



Learning non-linear patterns

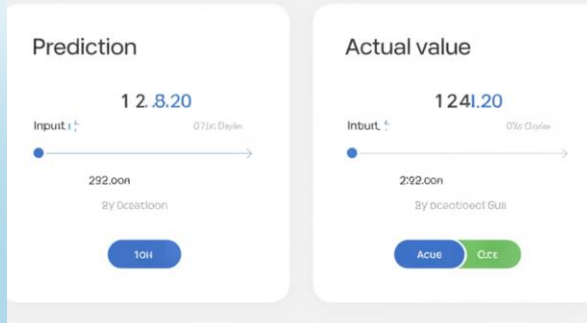


Layer collaboration



“Without activation functions, no matter how many layers we use, the neural network would behave just like a single-layer perceptron.”

How Neural Networks Learn



Learn iteratively by predicting and correcting errors

Process inputs through layers in a forward pass to generate outputs

How Neural Networks Learn

Steps Involved

Forward pass

Process input data through weighted layers to produce output predictions

Loss calculation

Quantify prediction errors using appropriate loss functions

Loss minimisation

Adjust network to reduce error and improve prediction accuracy

Understanding Backpropagation

Key Functions

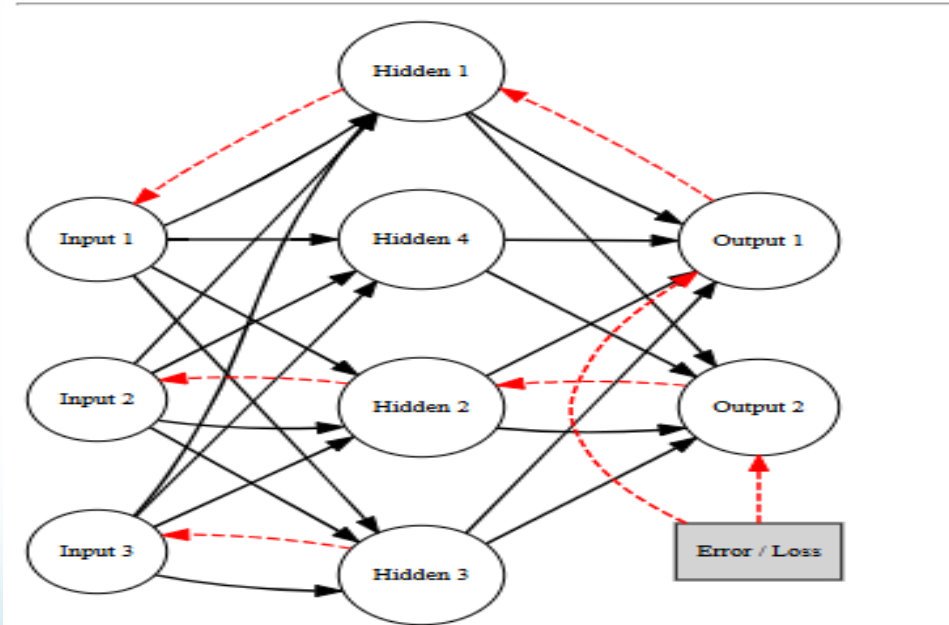
Backward pass

- Backpropagation calculates each weight's contribution to the error
- Uses gradients to determine the direction and magnitude of adjustments

Weight updates

- Optimiser (commonly Gradient Descent) updates weights to reduce loss
- Iteratively fine-tunes weights, improving prediction accuracy

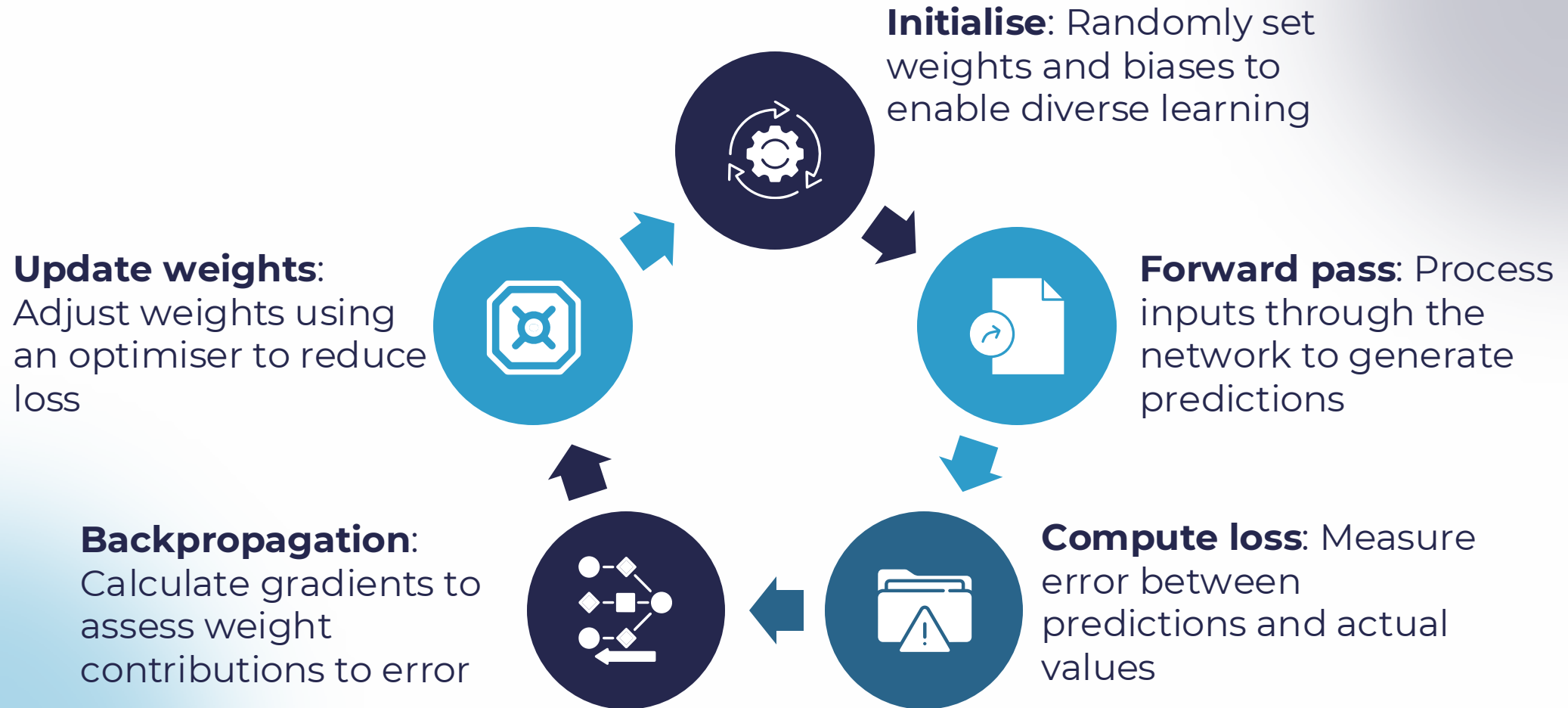
Understanding Backpropagation



Forward propagation (black) and backward propagation (colored) through a neural network

- Enables neural networks to learn from errors
- Drives continuous improvement through weight updates

The Complete Neural Network Training Cycle



The model iteratively trains over multiple epochs, gradually minimising loss and approaching optimal performance.

Reinforcement Learning

Reinforcement Learning

A Unique Branch of Machine Learning

Interactive learning

Agent learns optimal behaviour by interacting with the environment

Trial and error

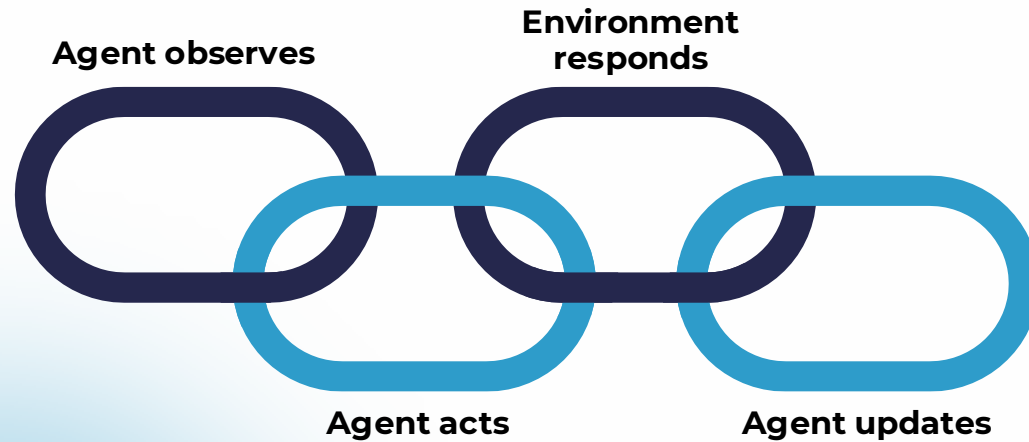
Learns through experimentation rather than explicit instructions

Reward-driven

Guides actions using reward signals to indicate success or failure

Reinforcement Learning mimics human and animal learning through feedback and consequences.

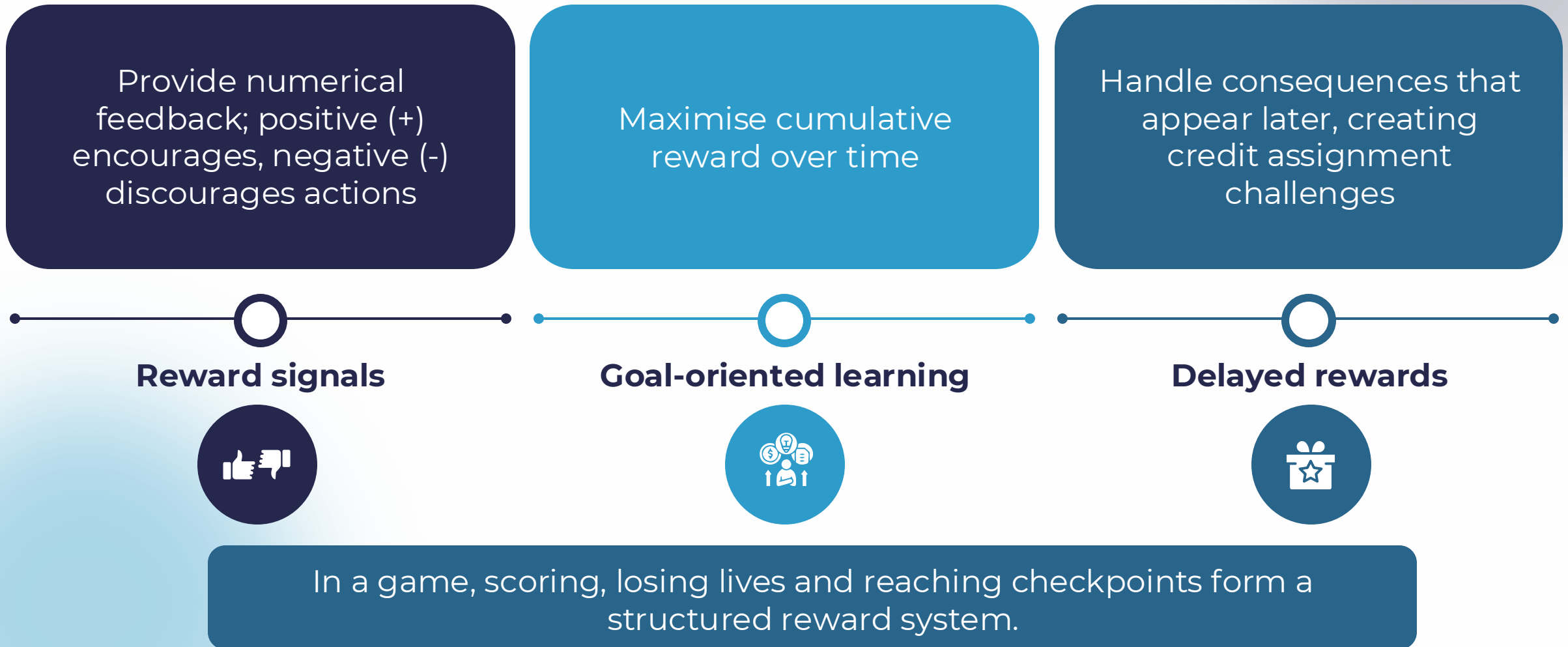
The Agent-Environment Loop



- **Agent:** observes states and selects actions
- **Environment:** responds with new states and feedback
- **Interaction:** cycles through observation, action and adaptation

The Power of Rewards

The Core Motivator in Reinforcement Learning



What Makes RL Unique

Beyond supervised learning

Operates without labelled data or explicit answers

Beyond unsupervised learning

Learns optimal strategies, not just patterns

Learning through experience

Develops policies via direct interaction

Real-World Applications of RL



Game AI

Chess, Go, Atari games,
poker



Robotics

Navigation, manipulation,
control



Business

Recommendations, pricing
and resource allocation

How These Concepts Lead to Agentic AI



AI

Powers intelligent systems like Siri, Google Translate and ChatGPT



Supervised & unsupervised learning

Enables machines to learn from data



Deep learning

Drives advances in vision, speech and NLP via neural networks



Reinforcement learning

Teaches agents optimal actions through trial, error and rewards

How These Concepts Lead to Agentic AI

Agentic AI = The Next Step



Integrates ML, DL and RL into autonomous agents



Enables agents to perceive, plan, decide and act in dynamic environments



Extends capabilities from answering questions to performing tasks autonomously

Demo: Google Teachable Machine
Refer to exercise document

Demo: Building a Conversational Chatbot with Transformers

Building Conversational Chatbot with Transformers

What Are We Doing?



Use Hugging Face Transformers to build a simple chatbot

Load a pre-trained conversational model (DialogPT)

Take user input, encode it, generate a response and maintain chat history

Demo: Language Translation with Transformers

Language Translation with Transformers

What are we doing?



Use Hugging Face Transformers pipeline API for language translation

Load a pre-trained translation model (Helsinki-NLP/opus-mt-en-fr)

Convert English text into French with minimal code

Q & A

Thank you